

MUHAMMAD SALMAN AHMAD

Software Engineer | Full-Stack & AI Engineer | Mobile, Backend & Cloud Architecture

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of full-time experience building and shipping production systems across mobile, web, backend, and cloud. Currently technical lead on a mentorship platform, owning product development end to end — architecture, full-stack engineering, UI/UX, AI integration, and technical decision-making. Experienced in event-driven cloud architecture on Firebase and Google Cloud, scalable REST APIs with Node.js and FastAPI, cross-platform apps in React Native and Flutter, and production AI systems including Retrieval-Augmented Generation and LLM-powered agents.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Dart, C++, SQL

Frontend & Mobile: React Native, React.js, Next.js, Flutter, Redux, TanStack Table, Tailwind CSS, Chart.js, Vite

Backend: Node.js, Express.js, FastAPI, Django / Django REST Framework, REST API design, WebSockets, microservices

Cloud & Data: Firebase (Firestore, Cloud Functions, FCM, Auth), Google Cloud Run, GCP, AWS S3 & EventBridge, PostgreSQL (Neon, pgvector), Upstash Redis, Docker, NGINX

AI & ML: RAG architecture, LangChain JS, Google Gemini embeddings, hybrid semantic + keyword search (RRF), LLM agents, CrewAI, prompt & context engineering

Practices: System architecture, event-driven design, technical leadership, UI/UX design, code review, Agile delivery

PROFESSIONAL EXPERIENCE

Software Engineer — RipeSeed

Jul 2026 – Present

Associate Software Engineer, Jun 2025 – Jul 2026 | Lahore, Pakistan

PRGRSS — Technical Lead / Product Engineer (Mentorship Platform)

- Own end-to-end product development across technical architecture, full-stack engineering, UI/UX, AI integration, and technical decision-making — working beyond feature delivery to identify product opportunities and design experiences that improve engagement and adoption.
- Lead feature development from product ideation and UI/UX design through system architecture, implementation, deployment, and maintenance.
- Design and maintain a scalable cloud architecture using React Native, Node.js/Express.js, Firebase Firestore, Firebase Cloud Functions, Firebase Cloud Messaging, and Google Cloud Run — with Firestore as the primary database, modeled around user, mentorship, social, and engagement workflows.
- Build and operate RESTful backend services on Cloud Run designed for independent deployment and horizontal scalability, with a focus on reliability, maintainability, and efficient data access.
- Implement event-driven workflows with Cloud Functions, including backend-triggered and asynchronous processes, plus a push-notification system over FCM delivering timely alerts for user activity, social interactions, and mentorship events.
- Designed and shipped the PRGRSS Social Feed, letting users share achievements, ask questions, and interact through likes and comments — adding an in-app community layer around the mentorship experience.
- Designed and implemented in-app subscription and monetization functionality, integrating payment infrastructure and supporting multiple membership and user-access models.
- Built and maintain the PRGRSS Admin Dashboard for user management, account review, report investigation, and platform oversight.
- Architect for long-term scalability across database design, API architecture, pagination, asynchronous processing, event-driven workflows, caching, and service boundaries.
- Currently developing an AI-powered mentorship-call agent that processes mentorship conversations and generates post-call insights — discussion summaries, action items, recommended next steps, and feedback — to help mentees get greater value from every session.
- Integrate AI capabilities into the existing product ecosystem with attention to data processing, backend orchestration, asynchronous workflows, and scalable API design.
- Own technical research, architectural decisions, production problem-solving, and third-party integrations, balancing product requirements, engineering quality, scalability, and delivery speed across mobile, backend, admin, cloud, and AI systems.

CALLI — Retrieval-Augmented Generation Platform

- Built and scaled a production-grade RAG platform with a Next.js chat interface and admin dashboard, powered by Google Gemini embeddings and Neon PostgreSQL.

- Engineered a hybrid search system combining semantic search (pgvector) and keyword search (tsvector) with Reciprocal Rank Fusion for materially improved retrieval accuracy.
- Designed an event-driven ingestion pipeline using AWS S3, EventBridge, and webhooks to automate document parsing, chunking, and indexing; tuned retrieval with LangChain JS and recursive character splitting for context-aware chunks.
- Developed a containerized FastAPI microservice (PyMuPDF, Pydantic) for asynchronous PDF processing and dynamic AI source highlighting.
- Integrated Upstash Redis as a distributed caching layer for embeddings and signed S3 URLs, significantly improving API response times and overall system performance.

Associate Software Engineer — TAG Solutions (TAGS)

Jan 2025 – Jun 2025

- Built scalable backend services and microservices with FastAPI.
- Designed and managed cloud infrastructure on Google Cloud Platform.
- Developed and maintained frontend web applications in React.js and .NET technologies.

Associate Software Engineer — CodeSuite

Jun 2024 – Jan 2025

- Built reusable component libraries shared across mobile and web applications.
- Developed responsive frontend interfaces in React.js and React Native, backed by RESTful APIs implemented in Django.
- Wrote clean, maintainable, testable code and participated in code reviews and technical discussions.

Co-Founder — Servizio.AI

Aug 2023 – Jan 2024

- Led AI-driven product innovation and technical direction for an early-stage venture.
- Built Flutter applications with polished UI, increasing user engagement by 40%.
- Introduced agile practices that cut development cycle time by 20% and lifted on-time delivery by 15%; instituted code reviews that reduced bug occurrences by 30%.

Software Engineering Intern — Arbisoft

Jul 2022 – Sep 2022

- Ranked 3rd in the 2022 internship program; built an intercity carpooling application end to end.
- Designed wireframes and UI in Adobe XD, modeled the relational schema, and developed the app with Flutter, Django, and Django REST Framework.
- Integrated RESTful APIs that improved app performance by 30% and reduced server response time by 50%; deployed via Docker containers for environment consistency.

SELECTED PROJECTS

Stoic — Automated Crypto Trading Platform (Backend & Infrastructure)

- Designed the system architecture for an automated trading platform built around Freqtrade as the core trading engine, integrated as an independent microservice.
- Implemented multi-tenancy by provisioning isolated Docker containers per user, each running a dedicated Freqtrade instance; configured NGINX as a reverse proxy for secure, efficient routing.
- Built secure, scalable APIs for user accounts, strategies, and trading sessions; deployed the full infrastructure on GCP Compute Engine.

WatchDogs — Analytics Dashboard

- Built a responsive React frontend on Vite with Chart.js visualizations and TanStack Table for sorting, pagination, and filtering over large datasets with a fully custom UI.

Deliveroo Clone — React Native

- Built a food-delivery app with Tailwind CSS styling, animated transitions, interactive map views via react-native-maps, and Redux-managed cart state.
- Engineered a Sanity CMS + GROQ backend that streamlined data querying and reduced server response times by 40%.

EDUCATION

BS, Computer Science — Air University, Islamabad

2020 – 2024

A-Levels, Computer Science — LACAS

2015 – 2019